INCH-POUND
MIL-DTL-32697
15 February 2022

DETAIL SPECIFICATION

CLOTH, NAPPED FRENCH TERRY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a napped French terry cloth. The fabric is available in one type.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY DEVCOM SOLDIER CENTER

Fabric Defect Replica Scales

(For information on obtaining copies of the Fabric Defect Replica Scales please contact U.S. Army DEVCOM Soldier Center, ATTN: FCDD-SCP-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

CODE OF FEDERAL REGULATIONS (CFR)

16 CFR 303 - Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at www.ecfr.gov.)

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8305

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U.S. DEPARTMENT OF AGRICULTURE (USDA)

United States Standards for Grades of Wool

(Copies of this document are available online at www.ams.usda.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC TM81 - pH of the Water- Extract from Wet Processed Textiles

AATCC TM135 - Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D1424	-	Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3787	-	Standard Test Method for Bursting Strength of Textiles – Constant-Rate-of-Traverse (CRT) Ball Burst Test
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM E96	-	Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM F1358/F1358M		Standard Test Method for Effects of Flame Impingement on Materials Used in Protective Clothing Not Designated Primarily for Flame Resistance

(Copies of these documents are available online at www.astm.org.)

CRC PRESS

Hayes, A. Wallace (editor) and Kruger, Claire L. (editor). *Hayes' Principles and Methods of Toxicology*. 2014.

(There are a number of ways through which this book may be obtained, such as online retailers, bookstores, and the public library, to name a few.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to a first article inspection (see 6.3) in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Standard sample. Unless otherwise specified (see 6.2), the finished fabric shall match the standard sample (see 6.4) for shade and appearance and shall be equal to or better than the standard sample with respect to all other characteristics for which the standard sample is referenced.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Basic material. The basic material for the knit base of the fabric shall be 100 percent wool.

3.5 Color. The basic material color shall be un-dyed natural.

3.6 Physical requirements. The fabric shall be a single-sided napped French terry made from 100 percent wool fiber and shall meet the requirements stated in table I when tested as specified in 4.4.

TABLE I. Physical requirements.

Characteristic	Requirement
Weight, ounces per square yard	10.5 – 12.5
Burst strength initial, pounds (minimum)	75
Break strength, pounds (minimum), elongation (percent)	
Wales	50, 75
Courses	65, 125
Tear strength, pounds (minimum)	
Wales	4.0
Courses	8.0
Moisture vapor transmission rate (MVTR), grams per square meter in 24 hours (minimum)	
Upright	1000
Inverted	3500
Dimensional stability, percent shrinkage (maximum)	
Wales	15
Courses	5
Effects of flame impingement	No melt/no drip
Grade (minimum)	60s

3.7 Finish. The fabric shall be napped.

3.8 pH. The pH of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.6.

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3.9 Toxicity. The finished fabric shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.4.7. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.10 Length, width, and put-up.

3.10.1 Length and put-up. Unless otherwise specified (see 6.2), for Government procurements only, the cloth shall be furnished in continuous lengths, each not less than 40 yards, and each length shall be put-up on a roll as specified (see 6.2) when examined as specified in 4.4.4.

3.10.2 Width. Unless otherwise specified (see 6.2), for Government procurements only, the width of the finished cloths shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.11 Roll identification.

3.11.1 Face identification. The napped side shall be identified as the face. The face side of the fabric shall be rolled inward.

3.11.2 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with 16 CFR 303 when examined as specified in 4.4.

3.12 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification, be uniform, and be free from defects specified in 4.4.5. The occurrence of defects shall not exceed the established Acceptance Quality Limits (AQLs) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified in the contract (see 6.2), a first article, submitted in accordance with 3.1.1, shall be examined and tested in accordance with 4.4.1 through 4.4.7 for compliance with the appearance, color, finish, design, workmanship, and testing requirements specified in this document.

4.2.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), a first article sample shall be provided. The sample unit shall be one continuous 5-yard length of full width cloth. The presence of excessive defects or failure of any testing shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests as specified in 4.4.1 through 4.4.7. The Government reserves the right to inspect and test all components, materials, and end items to determine conformance to stated requirements.

4.3.1 Conformance inspection samples. Sampling for conformance inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with AQLs as specified in the contract or order (see 6.2). The sample unit shall be one yard of finished cloth and the lot size shall be expressed in yards.

4.3.2 Material inspection. The wool shall be examined for grade in top form by comparison with the applicable U.S. Standard. The sample unit shall be 1 yard of undyed top. The sample size shall conform to that in 4.4.5. The lot size shall be expressed in units of one pound. The results from each sample unit shall be reported separately. The lot shall be unacceptable if a single sample unit fails to meet the grade requirement.

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4.4 Inspections. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.4.1 Visual inspection. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with MIL-STD-3064, type V. In the event of a conflict between the defects cited in MIL-STD-3064 and those in 4.4.5, this specification takes precedence.

4.4.2 Roll identification and marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.3 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.4 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.4.5 Classification of defects. The required yardage shall be examined for defects as listed in table II. The fabric shall be examined on the face side and the back side. The defects found shall be counted individually, regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each yard or fraction of a yard in which it occurs. The sample unit shall be 1 linear yard. The lot size shall be expressed in units of 1 yard each. The AQLs shall be as specified in the contract or order (see 6.2). The sample size and number of rolls shall be in accordance with table III.

TABLE II. Defects.

Defect Description	Points
Any hole, cut, tear, or abrasion mark.	4
Edge ravels when pulled outward.	4
Broken or missing yarn, dropped stitch, or run.	4
Skewing.	4
Any spot, stain, or streak more than 1 inch in combined directions.	4
Any mend.	4
Stretch (stitch tension), knitting tension too high or too loose, or washboards.	4
Coarse or fine yarns resulting in noticeable differences in appearance, stiffness, and thickness.	4
Twisted yarn.	4
Knots greater than Fabric Defect Replica Scale level D.	4
Slubs greater than Fabric Defect Replica Scale level F.	4
Width not as specified within a tolerance of not less than 1/2 inch.	4
Color not a good match to reference.	4
Objectionable odor.	4
Slack or tight selvages (when applicable).	4
Any embedded foreign matter or uncleanness.	4
Edges rolled, folded, creased, scalloped, or corded or baggy, ridgy, or wavy cloth.	4
Tackiness (sticky to touch).	4
Pile not napped.	4

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TABLE III. Physical requirements verification sampling.

Lot Size	Sample Size (# of Sample Units)
800 or less	2
801 through 22,000	3
22,001 and over	5

4.4.6 Physical requirements verification. Unless otherwise specified in the contract (see 6.2), the cloth shall be examined for conformance to the physical requirements as specified herein. The methods of testing shall be as specified in table IV.

TABLE IV. Physical requirements verification.

Characteristic	Requirement	Verification
Weight	Table I	ASTM D3776/D3776M
Burst strength initial	Table I	ASTM D3787
Break strength, elongation		
Wales	Table I	ASTM D1424
Courses	Table I	ASTM D1424
Tear strength		
Wales	Table I	ASTM D5034
Courses	Table I	ASTM D5034
MVTR		
Upright	Table I	ASTM E96, procedure B, 24 hours
Inverted	Table I	ASTM E96, procedure BW, 2 hours
Dimensional stability		
Wales	Table I	AATCC 135, 3, III, Aiii
Courses	Table I	AATCC 135, 3, III, Aiii
Effects of flame impingement	Table I	ASTM F1358
Grade	Table I	United States Standards for Grades of Wool
pH	3.8	AATCC TM81

4.4.6.1 Sampling and acceptance criteria for verification of physical requirements. Sample size shall be in accordance with table III. The sample unit shall be 5 continuous yards, full-width, of the finished cloth from different rolls of cloth in the sample. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified in this document.

4.4.7 Toxicity test. When specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the fabric is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see Hayes' Principles and Methods of Toxicology). If the toxicity requirement (see 3.9) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers napped French terry used as lining material for handwear by the United States Marine Corps.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. The specific issue of individual documents referenced (see 2.2 and 2.3).

c. Color required.

d. When first article sample is required (see 3.1.1, 4.2, and 6.3).

e. National stock number.

f. When conformance inspection is required (see 3.1.2).

g. Applicable Government standard samples, patterns, and drawings, including revisions (see 3.2).

h. Length, width, and put-up required (3.10.1 and 3.10.2).

i. AQLs required (see 3.12, 4.3.1, and 4.4.5).

j. Number of first article inspection samples (see 4.2.1).

k. Requirements and conditions for inspections (see 4.4).

l. Requirement for verification of physical requirements (see 4.4.6).

m. When toxicity testing is required (see 4.4.7).

n. Packaging requirements (see 5.1).

6.3 First article. When first article inspection (see 3.1.1) is required (see 6.2), the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Standard sample. For standard samples, address the procuring activity issuing the invitation for bids or request for proposal.

6.5 Sources of supply. A known source of 100 percent wool, napped French terry is style #2196, 11.5 (±0.5) ounces, available from Draper Knitting Company (www.draperknitting.com or 28 Draper Lane, Canton, MA 02021-1598).

6.6 Subject term (key word) listing.

Knit

Wool

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CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC
Air Force – 11

Review activities:
Army – AV, CR, IH, MI
Navy – AS, CG1, NU
Air Force – 03
DLA – CT

Civilian agency:
GSA – FAS

Preparing activity:
Navy − MC
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